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Artificial neural networks for predicting optical conversion efficiency in luminescent solar concentrators

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ABSTRACT

Developing light-harvesting materials able to shape the sunlight to cope with the absorption region of photovoltaic (PV) cells presents an opportunity for the utilization of spectral converters like the luminescent solar concentrators (LSCs). This study explores the use of artificial neural networks (ANNs) to predict the optical conversion efficiency of spectral converters, based on the material properties employed in their production, without the need for expensive and time-consuming experimental testing. To predict efficiency as a function of materials and manufacturing processes, ANNs were trained using data from previously documented physical implementations. The findings indicate that ANNs, having 97 and 19 neurons in the hidden layers, provide accurate efficiency predictions, making them a valuable tool for designing and optimizing spectral converting systems. The proposed model was validated and got a mean square error in the order of 10^{-5} for the optical conversion efficiency. The trained ANN introduced a novel methodology for predicting the optical efficiency of spectral converters, opening the door to the application of machine learning as a decision-making tool for material design, and eliminating the necessity for physical device implementations.

1. Introduction

The continuous demand for more energy-efficient cities and communities is fostering the search and development of strategies to achieve zero-energy buildings (ZEBs) [1], as buildings are one of the most energy-consuming sectors, along with the industry and transportation sectors [2,3]. The implementation of ZEBs requires an increase in energy efficiency and the use of renewable energies [1]. In this sense, the enhancement of photovoltaic (PV) devices and their integration in existing buildings is of prime importance, as the Sun is the dominant renewable energy source [4].

One of the major limitations of all PV devices, either being commercialized or still in an emerging state, is their inability to effectively collect all photons available in the solar spectrum. To tackle this issue, spectral converters can be applied, as they exploit the photoluminescence processes to capture low- or high-energy photons that cannot be used by the PV cell and convert them to photons with useful energy [5]. Those can be present as layers which are additive devices directly applied on the PV cell's surface to enhance PV performance (Fig. 1a), or luminescent solar concentrators (LSCs) which can be used in the built environment as they could be embedded in building façades or windows (Fig. 1b), allowing the generation of additional energy. LSCs consist of planar waveguides doped or coated with emissive materials, which absorb sunlight over a large area, which is converted and concentrated onto a smaller one, then directed to a solar cell, where it is converted into electrical energy [6]. The increasing interest in building-integrated PVs, as a solution where solar power generation can be incorporated into existing elements of the built environment, leverage research in the search for spectral converting solutions with high efficiencies [7], specially LSCs as they can be seen as one of the most promising technologies for semitransparent devices that can be integrated into the built environment without detrimental effects to the aesthetics of the building or the quality of life of the inhabitants, taking an important place towards the achievement of the so-called net zero-energy buildings [8–10]. However, predicting and optimizing their performance can be challenging due to various factors, such as the processing and optical properties of the luminescent materials and the geometry. To address these challenges, a multidisciplinary approach, incorporating expertise in materials science, optics, and PVs, is required.

Materials science and technology are on the verge of experiencing the impacts of digital transformation ahead of many other fields. Digital

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Nomenclature

Abbreviations

PV	Photovoltaic
LSC	Luminescent solar concentrator
ANN	Artificial neural network
ZEB	Zero-energy building
ML	Machine learning
PCE	Power conversion efficiency
PMMA	poly(methylmethacrylate)
SEBS	styrene-ethylene-butylene-styrene
PDMS	polydimethylsiloxane
PVP	polyvinylpyrrolidone
t-U(5000)	tri-ureasil organic–inorganic hybrid
d-U(600)	di-ureasil organic–inorganic hybrid
CD	Carbon dot
QD	Quantum dot
MSE	Mean squared error
MAE	Mean absolute error
Ln	Lanthanide

Machine learning symbols

$y_k^l(n)$	output signal of the k-th neuron in the l-th layer
$w_{k\nu}^l(n)$	weight between the l - layer k- neuron and the $(l-1)$ - layer ν- neuron
l	layer
k	k neuron
ν	ν neuron
α	Learning rate
$J\,()$	Cost function
θ	Parameters to be optimized
R^2	Coefficient of determination
n	Number of sample points
$\widehat{Y}$	Estimated value
Y	Measured/reported value
i	Input layer
h_1	First hidden layer
h_2	Second hidden layer
o	Output layer

Luminescent solar concentrators symbols

η_{opt}	optical conversion efficiency
η_{yield}	Absolute emission quantum yield
N_{Emi}	Number of emitted photons
P_{out}	Output optical power
N_{Abs}	Number of absorbed photons
P_{in}	Incident optical power
A_p	peak absorption wavelength
A_{min}	minimum wavelength value for the absorption spectral band
A_{max}	maximum wavelength value for the absorption spectral band
E_{min}	minimum wavelength value for the emission spectral band
E_p	wavelength value for the peak emission
E_{max}	maximum wavelength value for the emission spectral band

innovation has already introduced new categories of materials, thanks to its ability to enhance our predictive capabilities regarding material properties, the performance of active molecules, and even toxicological effects. To fully harness this potential, it's imperative to leverage artificial intelligence and machine learning (ML) effectively. The increasing importance of ML in materials science can be attributed to its proficiency in managing complex and extensive datasets. These algorithms can efficiently process and analyse such data, enabling researchers to identify patterns and relationships that would be difficult to detect using manual analysis. In materials science, ML has been used to predict material properties, optimize materials synthesis and processing, and accelerate the development of new materials [12].

Using machine learning techniques, Srivastava *et al.* [13] have addressed the challenge of composition-dependent degradation in hybrid organic–inorganic perovskites under environmental stress, which hampers their commercial viability. To circumvent the limitations of trial-and-error methods in assessing the behavior of these hybrids when exposed to different stressors, they have employed machine learning models with *in situ* photoluminescence data to predict the response of $Cs_yFA_{1-y}Pb(Br_xI_{1-x})_3$ to relative humidity cycles. Three models, including linear regression, echo state network [14], and seasonal autoregressive integrated moving average with exogenous regressor algorithms [15], have been quantitatively compared. The latter model achieves over 90 % accuracy in forecasting photoluminescence changes over a 50-hour period. Notably, samples with 17 % Cs content consistently demonstrate increased photoluminescence during the cycles. These precise time-series predictions can be extended to other families of hybrid organic–inorganic perovskites, highlighting the potential of ML approaches to development of materials for clean-energy applications.

Artificial neural networks (ANNs) are a type of ML algorithm inspired by the structure and function of biological neural networks in the brain and are capable of learning and generalizing from input-output data [16]. ANNs have been widely used as a powerful data-based alternative to physical modelling and predict the performance of complex problems, for example, to predict the performance of solar cells or other photonics devices [17]. ANNs are particularly suited to tasks such as image and speech recognition, where the input data is complex and high-dimensional, and their use in predicting spectral converting devices' efficiency is a new and active area of research [18]. One of the main advantages of employing ANN for the spectral converters design is the ability to process large datasets and identify complex non-linear relationships between the input and output variables, learn from large datasets, and capture subtle relationships between the input and output variables that may not be obvious from traditional models.

In recent years, ANN was used to predict the absorption and emission spectra of organic dyes [19], the photophysical properties of organic molecules with high accuracy [20], and the properties of quantum dots, including absorption and emission spectra, size distribution, and surface chemistry [21]. Similarly, a genetic algorithm was used to maximize the power conversion efficiency (PCE) of LSCs [22]. In addition, solar energy harvesting studies have used ANNs to predict distinct aspects related with solar energy, improving by 22 % the estimations using other prevision methods [23]. In particular, ANNs were used to predict the efficiency and power output of PV cells under different environmental conditions, including irradiance, temperature, and wind speed [24], to predict the performance of dye-sensitized solar cells [25] and to derive hourly global horizontal irradiance [23,26]. Another area of research in this solar energy field is the use of ANNs for PVs' fault diagnosis and maintenance including shading, module mismatch, and aging [27], to predict the required maintenance for cells based on their performance and environmental conditions [28]. Recently, ML clustering was used to optimize the design of solar cells demonstrating energy yields of more than 1 MW•h•m^{-2}•year^{-1} [29].

Chugh *et al.* [30] employed ANN to compute optical properties of solid-core photonic crystal fibers (PCFs), which included the effective index, effective mode area, dispersion, and confinement loss. The ANN model demonstrated high accuracy in predicting these properties within typical parameters, considering wavelength (0.5–1.8 μm), pitch (0.8–2.0 μm), diameter-to-pitch ratio (0.6–0.9), and the number of rings (4 or 5) in silica solid-core PCFs. The authors also showcased the efficiency of ANN for predicting optical properties with unknown device parameters, resulting in faster computational runtimes compared to traditional numerical simulation methods like the finite difference method [31], finite element method [32], block-iterative frequency-domain method [33], and plane wave expansion method [34,35]. Lo Brano *et al.* [36] focused on the development of a neural network prediction model to assess the energy producibility of dish-Stirling systems for electricity generation, offering a tool for system design and sizing across various installation sites. Their approach used real data for training, rather than predictions derived from analytical or numerical models. Various neural network configurations were explored, adjusting depth, neuron count, and computing resources for different input variables. The most successful network achieved a high coefficient of determination (0.98) when comparing predicted electrical output power to experimental measurements, demonstrating the reliability of these neural models.

Overall, ANNs have shown great promise for determining the photonic properties of materials and are increasingly being used in both academia and industry for material discovery, design, and analysis. They have shown great potential for designing and optimizing solar energy harvesting systems, enabling researchers to rapidly explore a large design space and identify optimal configurations for spectral converters that can lead to improved efficiency and performance. On the premise that spectral converters are devices that can effectively increase the efficiency of solar cells and/or concentrate incident solar radiation and that the optical efficiency of spectral converters is an important parameter in the design and optimization of these devices, this work explores a groundbreaking, multi-dimensional approach that leverages ANNs for predicting the performance of spectral converters, with a primary focus on LSCs. Importantly, this approach is equally applicable to converting layers since the predominant optical characteristics remain consistent, eliminating the need for physical device implementation. The option between ANNs among other ML methods was based on the specific problem, data availability, computational resources, and the research goals of our problem, where the complexity

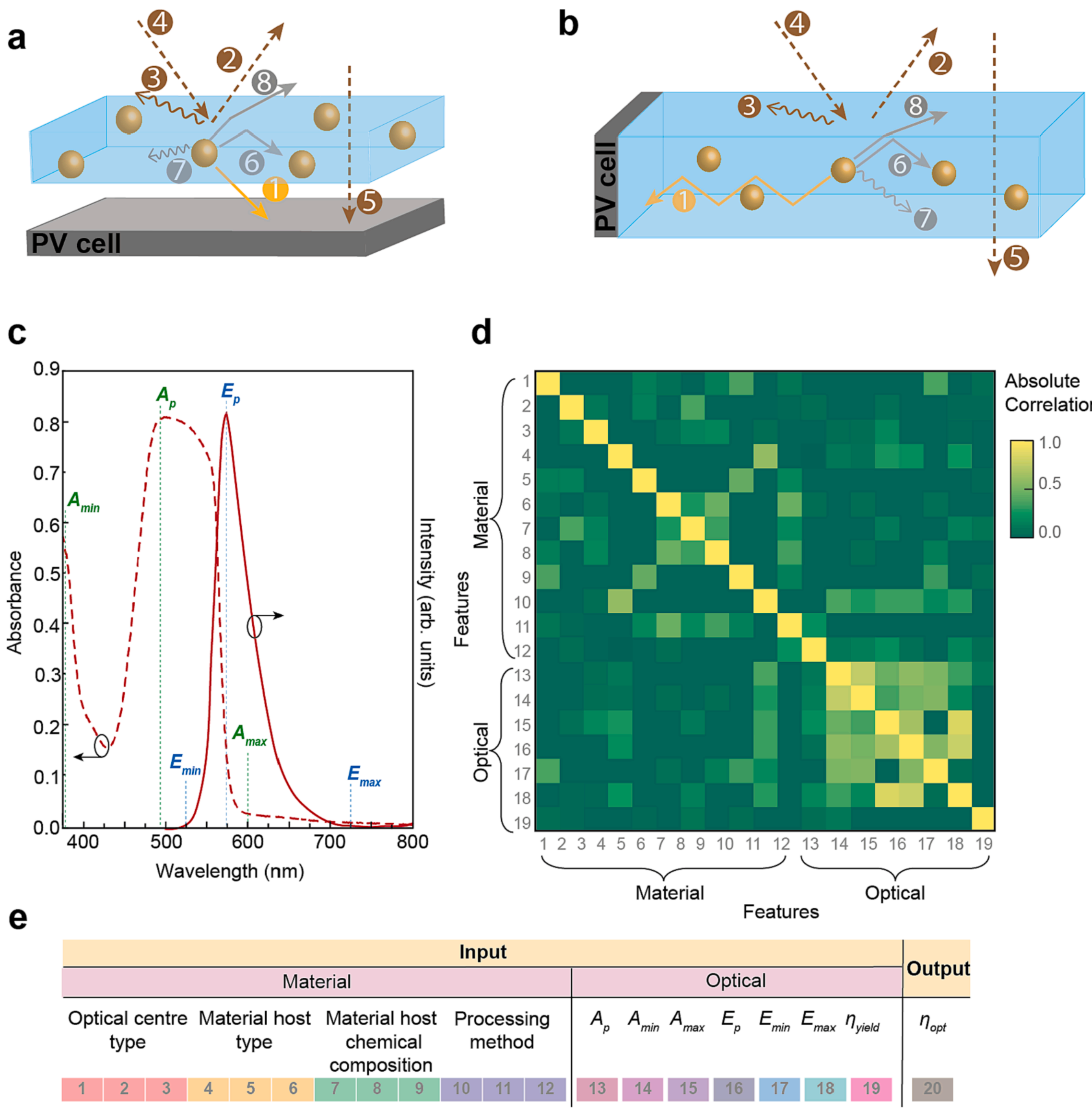

Fig. 1. Primary processes and losses occurring in (a) downshifting/downconverting and (b) planar LSCs: (1) emission from the optically active center, (2) Fresnel reflections for the incident sun light, (3) surface scattering, (4) incidente sun light, (5) direct transmitted radiation, (6) re-absorption by neighbor centers, (7) non-radiative relaxation, (8) emission within the escape cone [11]. c) Emission and absorption spectra of d-U(600) doped with Rhodamine 6G organic dye. d) Cross-correlation between the 19 features of the dataset, whose attribution is detailed on (e).

and adaptability of ANNs may offer a significant advantage [37].

While machine learning has found extensive application in the context of PV cells, there has been relatively limited research into its application in LSCs. Consequently, the present study contributes to advancing and accelerating this field, providing a valuable predictive model for optical efficiencies by considering intrinsic photophysical properties (absorption, emission, and optical center type) and material characteristics (chemical host and processing). The potential of the results achieved here will allow us to obtain a simplified optimized design for the various parameters affecting the spectral converter devices, with the potential to propel the overall field of solar energy industry into a new era of enhanced energy capture and utilization.

2. Methodology

2.1. Artificial neural networks (ANNs)

ANNs are machine learning algorithms inspired by the structure and function of biological neural networks, used to model complex non-linear relationships between inputs and outputs, and can be highly effective in solving a wide range of problems. Essentially, the structure of a typical ANN consists of neurons, which are processing elements, and links, which are interconnections between the neurons, controlled by a weight parameter that determines the strength of the connection. When a neuron receives stimulus from other neurons, it processes the information and produces an output. In this study, we employ a feedforward ANN with two hidden layers to model the correlation between the crucial parameters of LSCs and their efficiency. The number of neurons in the output layer equals the number of output variables, while the input nodes correspond to the characteristics of the LSC material, as well as the wavelength of both the incident and emitted light. The output layer of the network consists of one neuron, representing the predicted output.

The activation function used for all neurons in this implementation of the network non-linearity is the sigmoid function, which computes the output signal of the k-th neuron in the l-th layer as follows [38]:

$$y_k^l(n) = \frac{1}{1 + e^{\sum_{\nu=0}^{p_{l-1}} w_{k\nu}^l(n)}} \tag{1}$$

where $w_{k\nu}^l(n)$ is the weight between the l- layer k- neuron and the $(l-1)$ -layer ν- neuron.

Each neuron applies an activation function to the weighted sum of its inputs, producing an output value. The network is trained by adjusting the weights and activation functions through a process called back-propagation [39]. Backpropagation tries to minimize the error between the predicted output and the actual output, allowing the weights and activation functions to be adjusted iteratively using optimization algorithms like gradient descent [40], by propagating the error back through the network and updating the weights accordingly.

2.2. Data source

The optical conversion efficiency (η_{opt}) was chosen to be the predicted output parameter, as it is the most common figure of merit used to describe the performance of LSCs (Supplementary Information for details). The dataset [41] (https://figshare.com/s/23186603367d9c9483c0) used to train and test the proposed algorithm was obtained from a public repository provided by an academic institution. The selected data corresponds to 56 % of the total number of entries available on the dataset, and only the features related to the type of optically active center and host materials, processing method and respective absorption and emission properties (spectral regions and emission quantum yield, η_{yield}) were considered, as well as the correspondent reported η_{opt} values (Table S1, Supplementary Information).

A preliminary analysis was conducted to ensure that the dataset used in this study was sufficiently large to capture the complexity of the problem and include a diverse range of examples, which would enable the model to generalize well to new data. Real-world data often comes with missing attributes or values related to the measured optical properties. Therefore, it is highly beneficial to pre-process the dataset by cleaning and organizing the raw data and ensuring the accuracy, diversity, and representativeness of the data in the problem domain before using it in the proposed ANN. In this case, the rows in the dataset with missing data for relevant features were eliminated, resulting in a dataset of 236 entries (Tables S1 and S2, Supplementary Information).

Eighteen-point two percent (18.2 %) of the dataset was reserved as the hold-up test set, comprising 43 examples (Table S2). This test set was curated using data from our previously published results, which encompass a wide spectrum of conversion efficiencies, enhancing our confidence in the utilized values. The remaining 193 examples (Table S1) were utilized for feature selection and model validation. We implemented a 10-fold cross-validation after executing the standard dataset split routine to determine the optimal ANN architecture and hyperparameters.

The next step involved converting the categorical classifiers to numeric ones using one-hot encoding. This process resulted in three features for each classifier. Additionally, we scaled the numerical values to fit within the entire operating range, specifically between 0 and 1. The spectral wavelengths were expressed in μm units, while the emission quantum yield η_{yield} (Eq. S4) and efficiency η_{opt} (Eq. S1) were defined as the ratios between the output (N_{Emi} and P_{out}, respectively) and the input (N_{Abs} and P_{in}, respectively). The categorical classifiers considered in this model were based on the manufacturing processing and included: i) optical center type, ii) material host type, iii) material host chemical composition, and iv) processing method. The classes for each of these classifiers are detailed in Table 1.

The numerical values considered for the optical characterization parameters include: i) peak absorption wavelength (A_p), ii) minimum wavelength value for the absorption spectral band (A_{min}), iii) maximum wavelength value for the absorption spectral band (A_{max}), iv) wavelength value for the peak emission (E_p), v) minimum wavelength value for the emission spectral band (E_{min}), vi) maximum wavelength value for the emission spectral band (E_{max}), and vii) absolute emission quantum yield (η_{yield}). To illustrate the physical meaning of each experimental parameter, Fig. 1c illustrates the absorption and emission spectra of the material with the largest experimental values for η_{opt} [42]. To gain insights into the relationships between multiple attributes in the dataset, we conducted a cross-correlation analysis between the 19 features, as depicted in Fig. 1d. In total, there are 19 input features (attributes) for one output corresponding to the LSC optical efficiency, resulting in a total of 31 descriptors. The final structure of the dataset features can be found in Figure S1, Supplementary Information and in Fig. 1e. This analysis is a valuable tool for identifying relationships between features in a dataset, even if they are not linearly related as it can reveal whether

Table 1
Classes for the categorical classifiers. Taken from [41].

Optical center type	Material host type	Material host chemical composition	Processing method
Quantum Dots	Polymer	PMMA	Waveguide
Organic Dye	Hybrid	SEBS	Films
Carbon Dot	Solution	PDMS	Bulk
Lanthanide	Other	PVP	Solution
Polymer		Polyurethane	Fiber
Other		t-U(5000)	Other
		d-U(600)	
		Other	

PMMA = poly(methylmethacrylate); SEBS = styrene-ethylene-butylene-styrene; PDMS = polydimethylsiloxane; PVP = polyvinylpyrrolidone; t-U(5000) = tri-ureasil organic–inorganic hybrid; d-U(600) = di-ureasil organic–inorganic hybrid.

one or more attributes are associated with other attributes.

Organic molecules and composites like dyes, CDs, and QDs have a Stokes shift (the energy difference between absorption and emission peaks) of approximately 4,200 ± 3,400 cm^{-1} (based on data from Table S1). On the other hand, lanthanide (Ln)-based compounds (identified as optical center type: Ln in Table S1) exhibit a much larger Stokes' shift, around 13,500 ± 4,800 cm^{-1}, often referred to pseudo-Stokes shift [43,44] because the photophysics process of these compounds involves two centers (one for absorption and one for emission). In the case of Ln-based compounds, absorption occurs at the organic ligands, typically in the UV spectral region, while emissions originate from the Ln ion in the visible to infrared spectral regions. This process is known as the antenna effect, where intramolecular energy transfer between ligands and Ln ions plays a fundamental role [44,45].

The results showed a strong correlation among optical features (entries 13 – 19 in Fig. 1e), particularly between E_{max} (entry 18 in Fig. 1e), E_p (entry 16 in Fig. 1e), and A_{max} (entry 15 in Fig. 1e). These findings are not surprising, as the emission and absorption wavelengths in LSCs are closely tied to the optical properties of the luminescent material employed. The spectral bands, whether for emission or absorption, may be typically characterized by a combination of Gaussian bands. Consequently, a correlation exists between the E_{max} and E_p within the emission band. However, an exception is η_{yield}, which shows weak correlations with all other features, regardless of the material and optical characteristics. This underscores the importance of using models like ANN to predict the non-linear relationship for the η_{opt} of LSCs, as it is proportional to η_{yield} (Eq. S3).

2.3. Algorithm

To evaluate the algorithm's ability to handle the challenge of unbiased inputs, the dataset was randomly shuffled and divided into 10 equal parts for cross-validation. The k-fold cross validation is a resampling method that allows reducing the bias of the model, considering k = 10 has been demonstrated to perform well relative to estimating bias and prediction error [46]. The hold-out sample was then used for testing the best method with optimized hyperparameters. As mentioned before, the hold-up sample was not used during training or validation and was used solely to assess the performance of the final model and was extracted from the original dataset, corresponding to values previously published by some of us, ensuring a thorough understanding of the parameters and errors associated with the experimental measurements. This step was crucial to ensure that the model could perform well on completely unseen data and was not biased toward the training or validation data.

Hyperparameter optimization was conducted during the cross-validation phase to choose the best model with optimal hyperparameters. The objective of this optimization was to select the best hyperparameters using the mini-batch gradient descent optimization algorithm, which is described as follows [47]:

$$\theta_{i+1} = \theta_i - \alpha \nabla_\theta J\left(\theta; x^{(i:i+n)}; y^{(i:i+n)}\right) \tag{2}$$

where α is the learning rate, $J(.)$ is the cost function, and θ are the parameters to be optimized. Noticeably, these parameters are not learned from the data but are set before training the model, such as the number of hidden layers, neurons in each layer, and learning rate. The same optimization algorithm was also used in the training process, where the initiated values for the weights, $w^l_{k\nu}(n)$, were random shoots. Only after the best-performing features were selected and the model was fully optimized, it was then trained using 82.8 % of the dataset reserved for train and tested on the remaining hold-out test set (18.2 %).

In the hyperparameter optimization and training the best parameters were estimated using the mean square error between the output values and the reported values of the LSC optical efficiency, considered as the cost function. The final accuracy of the LSC estimates was assessed using the mean squared error (MSE), the mean absolute error (MAE), and the coefficient of determination (R^2) statistics, defined as [48]:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \widehat{Y}_i)^2 \tag{3}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|Y_i - \widehat{Y}_i| \tag{4}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(Y_i - \widehat{Y}_i)^2}{\sum_{i=1}^{n}\left(Y_i - \frac{1}{n}\sum_{i=1}^{n}Y_i\right)^2} \tag{5}$$

where n is the number of sample points, $\widehat{Y}$ is the LSC estimates, and Y is the corresponding measured/reported values. The illustrative workflow for the employed procedure is illustrated in Fig. 2a. The optimized ANN, represented in Fig. 2b, has 19 input neurons, 97 neurons in the first hidden layer, 19 neurons in the second hidden layer, and 1 output neuron, for a learning rate of 0.001.

The second hidden layer consists of 19 neurons, with each neuron in this layer also receiving inputs from all the neurons in the previous layer (hidden layer 1) and using the same activation function as the first hidden layer. After the hyperparameter optimization and k-fold test, the complete dataset, excluding the hold-out samples, was used to train the neural network. The process involved feeding the input data to the network and adjusting the weights to minimize the error between the network output and the actual output. The training process was stopped either when the network performance on the validation set stopped improving with a cost function error smaller than 0.0001 (recall that the typical experimental accuracy for the reported η_{opt} values is 0.001), or when the number of iterations reached a pre-defined value, which in this case was 250 million.

Fig. 3a illustrates the evolution of the cost function value during the training process. As the maximum number of iterations was approached, the improvement in error became smaller, with an improvement of less than 10^{-9} for every 10^5 iterations.

After completing the training, we compared the predicted η_{opt} values with the actual values by creating a scatter plot, which is depicted in Fig. 3b. Our ANN model proved to be highly accurate in predicting η_{opt}, as evidenced by the low MSE = 5.5 × 10^{-3} when compared to the experimental uncertainty associated with this parameter measurement. The model also demonstrated robustness in predicting even high values. Additionally, the linear fit in Fig. 3b, with a slope of 1.0027 ± 0.0068 (R^2 = 0.997), nearly aligns with the $y = x$ trend, indicating minimal bias. In addition, it is worth noting that the data points with the worst prediction accuracy tend to cluster around 10 %. This can be attributed to potential misclassification of the original dataset or experimental anomalies/errors that could have influenced the reported measurement. As the number of outliers is reduced, the overall impact on the final results is minimal, the decision has been made not to take any further actions or dataset truncation.

3. Results and discussion

Once the ANN has been trained and evaluated, it is crucial to validate its performance on a new dataset to ensure its robustness and ability to generalize to unseen data. In this study, we utilized the hold-out testing dataset to evaluate the model's generalization ability, and the results are presented in Fig. 4. Table 2 summarizes the performance metrics of the model comparing the hold-up test and the 10-fold cross-validation average values.

Based on the three metrics indicating model performance, our model has achieved excellent accuracy. In conclusion, despite the relatively limited size of our dataset resulting from a scarcity of available reported values, we can confidently state that our model accurately predicts all reported values of η_{opt}, with uncertainties considerably smaller than

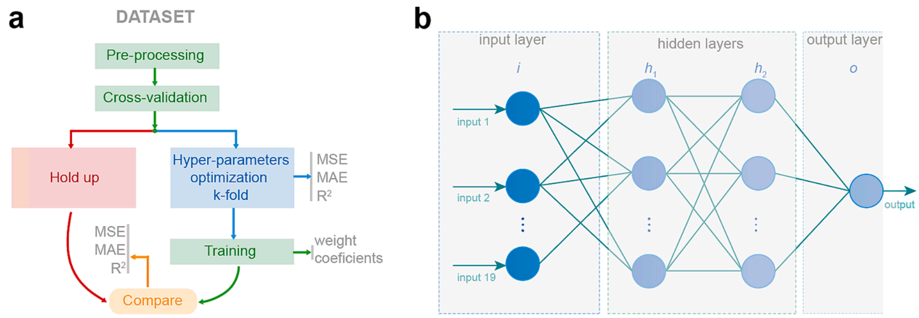

Fig. 2. (a) Structure of the ANN with 2 hidden layers, 19 neurons in the input layer (i), 97 neurons in the first hidden layer (h_1), 19 neurons in the second hidden layer (h_2) and 1 neuron in the output layer (o). b) Workflow describing the process used to predict the optical efficiency of LSCs.

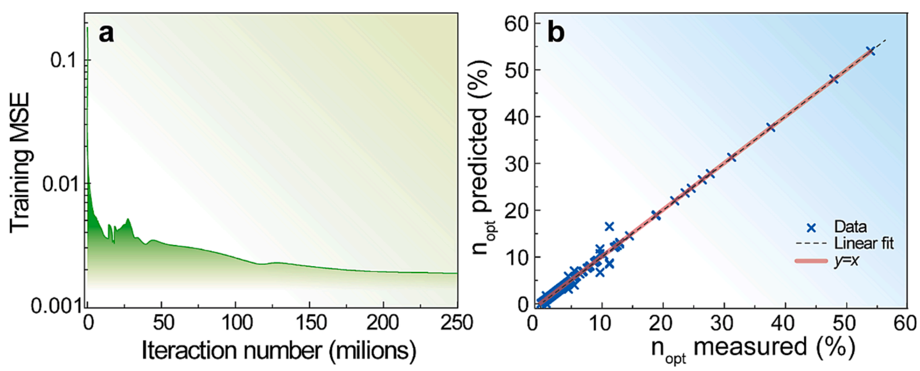

Fig. 3. A) MSE evolution along the training process. b) Predicted η_{opt} as a function of the measured values, after training and for the training dataset.

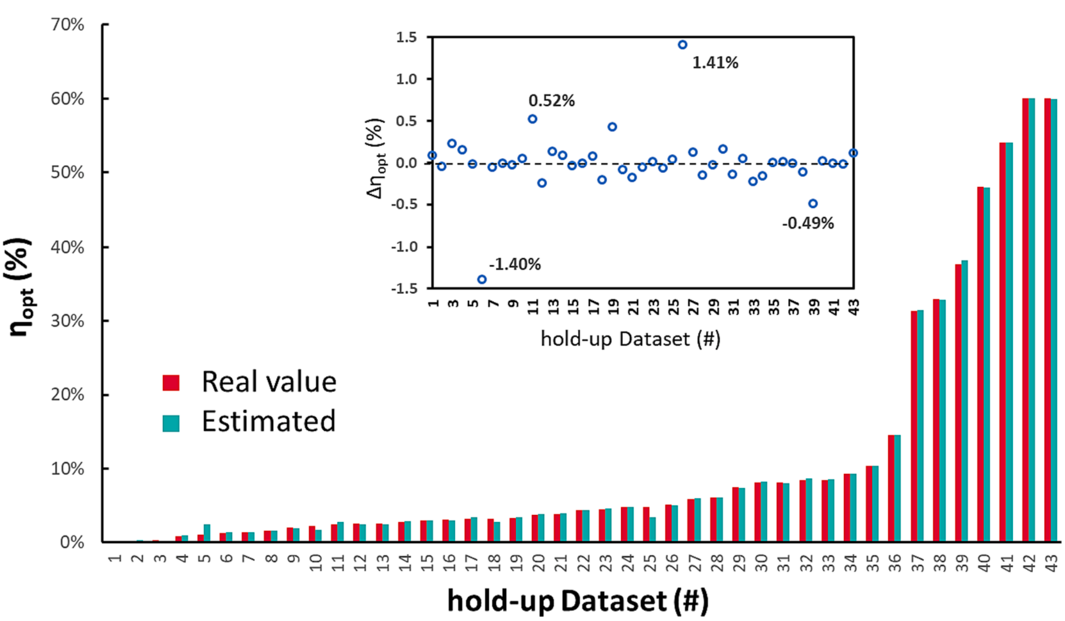

Fig. 4. Estimated from the ANN model and real values for the η_{opt} parameter for the hold-up dataset. See Table S2 for further details of each component of the hold-up test set. The inset shows the deviation between real and estimated values ($\Delta\eta_{opt}$).

those typically encountered during the experimental characterization of LSC. This study represents the first application of the ML method for predicting LSC performance, making it challenging to establish a benchmark comparison with other methodologies.

The hold-up test set involves a wide different parameters value, resulting in η_{opt} values that could vary by more than 270 times. Also, the ANN model demonstrates remarkable robustness by accurately predicting η_{opt} in systems within the same class of compounds, host material, and similar absorption features but differ in the type of Ln ion as the emissive center. For instance, a Tb^{3+}-β-diketone tetrakis complex Q[Tb(tta)$_4$], where Q = 1-butyl-3-[3-(trimethoxysilyl)propyl]imidazolium (entry #3 in Table S2) [49], consists of a coordination compound with four equal coordinating ligands and a counterion in the outer coordination sphere to balance the charge. The LSC based on this complex presents η_{opt} = 0.27%, while a LSC containing Eu^{3+}-β-diketone tris complex [Eu(hfac)$_3$•DPEPO], where hfac = hexafluoroacetylacetone

Table 2
Performance metrics for the ANN model. The uncertainty values for the cross-validation results were considered twice the standard deviation of the mean (2σ).

Performance metric	10-fold cross-validation	Hold up test
R^2	0.96893 ± 0.09722	0.99955
MSE	0.0055 ± 0.0207	1.2×10^{-5}
MAE	0.2230 ± 0.5750	0.0017

and DPEPO = bis(2-(diphenylphosphino)phenyl) ether oxide (entry #42 in Table S2), has η_{opt} = 60 % [50]. Both complexes share similar absorption features from β-diketone ligands and are hosted in the same material (PMMA). However, the ANN model could distinguish between them within errors of less than 1 %.

Another example involves the different preparation of LSCs with similar optical features but distinct material characteristics. For instance, consider carbon dots (CDs) based on Rhodamine B (entry #25 in Table S2) [51] and CDs based on citric acid (entry #35 in Table S2) [52]. Despite having similar optical features in terms of peak positions, they were prepared using different methods. The Rhodamine-based CDs were processed as a film in PVP, while the citric acid-based ones as bulk material in PMMA. Their η_{opt} values differ from 4.8% to 9.3%, respectively, demonstrating that the ANN model also captures the influence of experimental conditions. This highlights the ability of the model to infer η_{opt} even when the host material and processing methods are altered. These comparative examples validate how the ANN model developed here can be useful in the design of new LSCs.

The effectiveness of our predictions in this study can be credited to the careful selection of input features, as illustrated in Fig. S1. These features display non-linear relationship with the output variable, η_{opt}, underscoring the reliability of spectroscopic properties and material composition as descriptive factors for the optical conversion efficiency of LSCs.

4. Conclusion

Spectral converting devices are an innovative technology with the potential to significantly enhance solar energy conversion efficiency, but their design and optimization are challenging due to the interplay of various factors affecting their performance. However, determining the properties of such devices takes considerable time and effort. To address this challenge, we developed an ANN model as a tool for predicting the optical conversion efficiency (η_{opt}), considering the composition, structural, and optical characterization features of luminescent solar concentrators (LSCs). We evaluated the model using a dataset of 236 samples from a public repository, and the results show that the ANN model accurately predicts η_{opt} values. The model was trained using ~82 % of the total dataset, while the remaining ~18% was used for validation. The validation results revealed that the ANN could predict η_{opt} values with a mean squared error of 1.2×10^{-5}, revelling high accuracy and excellent generalization ability. The significance of this study lies in its ability to accelerate research in this field and contribute to the development of more efficient and sustainable solar energy technologies. The trained network parameters are readily available for use by stakeholders in the field (Supplementary Information). By using ANNs to model the complex relationship between various parameters and the spectral converters' performance, we can predict the efficiency of new devices without extensive experimental measurements. Therefore, this approach has the potential to accelerate the development of efficient systems and help transition to a sustainable energy future.

Author contribution

RASF and PSA contributed to research plan determination, acquisition of all data, data analysis, result discussion, and manuscript writing. SFHC contributed to data acquisition, data analysis, and manuscript writing. ANCN contributed to the manuscript revision and spectral analysis. PSA and LMSD defined the ANN.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.solener.2023.112290.

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